

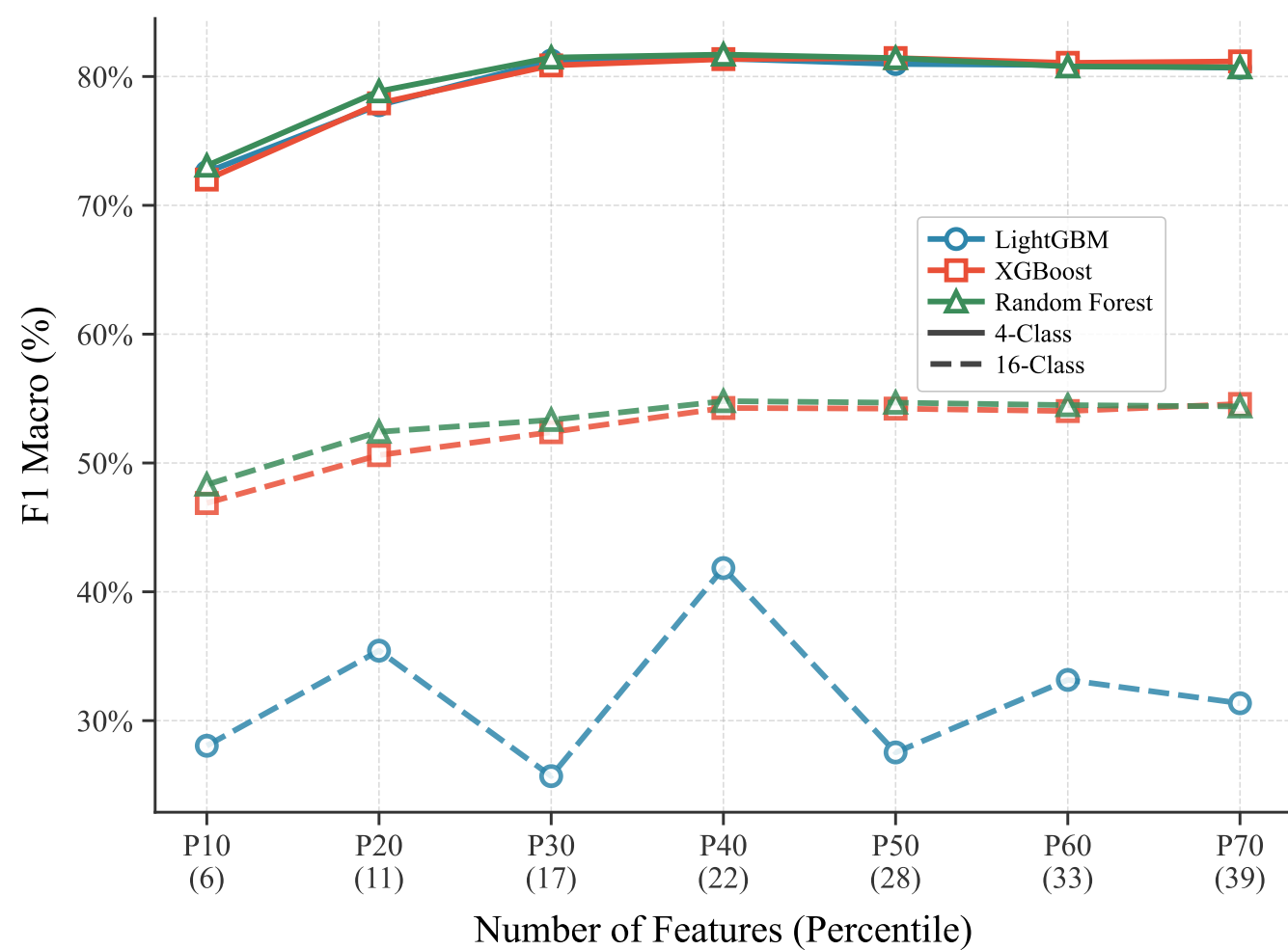
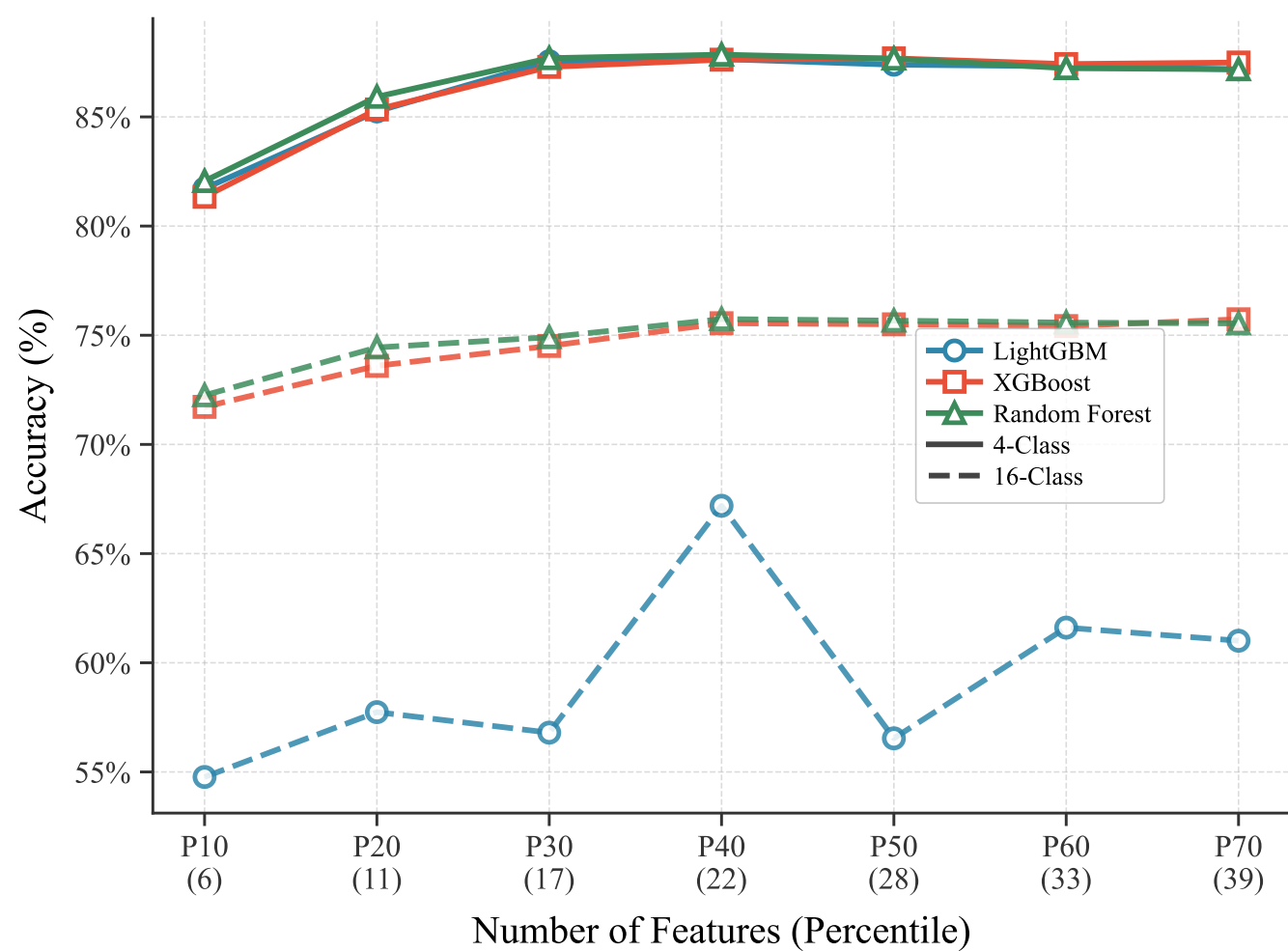


Figure 3. Accuracy (left) and macro F1-score (right) of LightGBM, XGBoost, and Random Forest under the 4-class (solid line) and 16-class (dashed line) classification scenarios, across feature subsets selected at increasing percentile thresholds (P10-P70); numbers in parentheses indicate the corresponding number of selected features.